

Investigating and Mitigating Biases in Crowdsourced Data

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ABSTRACT

It is common practice for machine learning systems to rely on crowdsourced label data for training and evaluation. It is also well-known that biases present in the label data can induce biases in the trained models. Biases may be introduced by the mechanisms used for deciding what data should/could be labelled or by the mechanisms employed to obtain the labels. Various approaches have been proposed to detect and correct biases once the label dataset has been constructed. However, proactively reducing biases during the data labelling phase and ensuring data fairness could be more economical compared to post-processing bias mitigation approaches. In this workshop, we aim to foster discussion on ongoing research around biases in crowdsourced data and to identify future research directions to detect, quantify and mitigate biases before, during and after the labelling process such that both task requesters and crowd workers can benefit. We will explore how specific crowdsourcing workflows, worker attributes, and work practices contribute to biases in the labelled data; how to quantify and mitigate biases as part of the labelling process; and how such mitigation approaches may impact workers and the crowdsourcing ecosystem. The outcome of the workshop will include a collaborative publication of a research agenda to improve or develop novel methods relating to crowdsourcing tools, processes and work practices to address biases in crowdsourced data. We also plan to run a Crowd Bias Challenge prior to the workshop, where participants will be asked to collect labels for a given dataset while minimising potential biases.

CCS CONCEPTS

• **Human-centered computing** → **Collaborative and social computing**.

KEYWORDS

crowdsourcing, data quality, biases

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1 INTRODUCTION

While quality assurance methods are commonly used with crowdsourced data labelling, studies have shown that annotator biases often creep into labelling decisions [6, 7, 18, 31, 34]. For instance, Hube et al., [18] show that workers with strong opinions produce biased annotations, even among experienced workers. More generally, there can be different types of biases in data (e.g., population bias, behavioural bias, temporal bias) [27, 28, 35]. Mehrabi et al. [27] provide an extensive list of 23 different types of biases in data. There are several methods available to account for such biases at different stages of data preparation and model training. However, there can be economic reasons and problem-specific needs that would make certain techniques more suitable for a particular stage.

Recent work has explored promising directions that can ensure data fairness when using crowdsourcing to collect data [4, 8, 9, 18]. Instead of presenting the task to the generic worker pool, one approach is to filter the workers based on attributes that can induce bias in the labels. For example, when gathering crowd labels, it is possible to mitigate bias by using a balanced or skewed sample of workers with respect to worker demographics (e.g., Age, Gender) and the minimum wage in their country [4]. Barbosa & Chen [4] allow requesters to decide how they want to manage the worker population. For example, when collecting audio data to train a voice assistant, requesters can set the worker pool to be diverse in terms of gender, age, native language. Similarly, we can also assign questions to specific workers or worker groups taking fairness into account in addition to budget constraints and overall accuracy [9]. In addition, task presentation strategies can help in easing bias. For example, social projection, awareness reminders and personalised alerts can reduce worker bias [18].

After the data collection process, bias can also be reduced using many other approaches such as data aggregation techniques [19], systematically adding new data points to fix coverage [2], and

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crowdsourced bias detection [17]. Other approaches to mitigate bias can be employed during feature engineering and model training. Prior work highlights the direct use of crowdsourced data [3], using crowds to identify perceived fairness of features [37, 38], using pre-processing methods (removing sensitive attributes, resampling the data to remove discrimination, iteratively adjust training sample weights of data from sensitive groups) [5, 20, 23] and using active learning [1]. However, techniques applied after data collection process can result in wasted effort. Furthermore, biases can be introduced as a result of many other factors such as poor task design [6], task attributes [19], due to sampling decisions the task requester makes about the data, and worker task selection preferences. Therefore, tackling biases during data collection is necessary as it may not be possible to rectify certain issues post-hoc.

Understanding and mitigating biases in crowd data is highly relevant to CSCW researchers and others who rely on crowd data for creating automated systems. In addition, researchers increasingly use crowdsourcing platforms to gather research data through survey tasks and user experiments. Several recent workshops such as Subjectivity, Ambiguity and Disagreement (SAD) in Crowdsourcing at WebConf 2019¹, Crowd Bias workshop at HCOMP 2018², Crowd Science at NeurIPS 2020³, and Data Excellence at HCOMP 2020⁴ have explored related topics with a particular focus on data, evaluation and applications. While continuing the broader discussion on biases in crowd data, in this workshop, we aim to focus on crowd workers and the crowdsourcing process. Therefore, we anticipate this workshop and its outcomes will be relevant and important to the broader CSCW community.

2 WORKSHOP

2.1 Goals and Themes

Through this workshop, we aim to foster discussion on ongoing work around biases in crowd data, provide a central platform to revisit the current research, and identify future research directions that are beneficial to both task requesters and crowd workers. We have identified the following four workshop themes that will focus on understanding, quantifying and mitigating biases in crowd data while exploring their impact on crowd workers.

1. *Understanding how annotator attributes contribute to biases*: Research on crowd work has often focused on task accuracy whereas other factors such as biases in data have received limited attention. We are interested in reviewing existing approaches and discussing ongoing work that helps us better understand annotation attributes contributing to biases.

2. *Quantifying bias in annotated data*: An important step towards bias mitigation is detecting such biases and measuring the extent of biases in data. We seek to discuss different methods, metrics and challenges in quantifying biases, particularly in crowdsourced data. Further, we are interested in ways of comparing biases across different samples and investigating if specific biases are task-specific or task-independent.

3. *Novel approaches to mitigate crowd bias*: We plan to explore novel methods that aim to reduce biases in crowd annotation in particular. Current approaches range from worker pre-selection, improving task presentation and dynamic task assignment. We seek to discuss shortcomings and limitations of existing and ongoing approaches and ideate future directions.

4. *Impact on crowd workers*: We want to explore how bias identification and mitigation strategies can impact the actual workers, positively or negatively. For example, workers in certain groups may face increased competition and lack of task availability. Collecting worker attributes and profiling could raise ethical concerns.

2.2 Structure and Activities

We have planned the workshop structure with a specific focus on virtual participation. Considering the potential spread of participants across multiple time zones, we plan on running a 5-hour synchronous workshop and an optional workshop challenge as a pre-workshop activity. In addition, we will invite workshop participants to submit position papers relating to workshop themes.

2.3 Pre-workshop Activities

Crowd Bias Challenge: To maximise the networking opportunities and participant engagement, we plan to introduce a workshop challenge (e.g., TREC Challenges) where participants will gather a crowdsourced dataset for a given problem while minimising potential biases in data. The challenge will start three weeks before the conference, and participants will be invited to attempt the challenge as individuals or in groups of up to 4 members. We will use a milestone and result-oriented leaderboard to motivate the teams.

The challenge will introduce a problem where biases in data are problematic (e.g., crowd judgements on content moderation). We will be providing the seed dataset for the participants to gather labels. To set up the crowd task and gather data, the teams will be provided with access to a crowdsourcing platform with a limited amount of credits (e.g., Amazon MTurk). The teams will then submit both original crowd labels and the aggregated result that they come up with. The results will be evaluated using pre-specified ground truth data and content categories that are susceptible to biases. For instance, we will consider the variation in accuracy across content categories in addition to overall task accuracy.

Position Papers: We will invite the participants to submit 2-page position papers on previous or ongoing research work on biases in crowd data.

2.4 Synchronous Workshop

The synchronous workshop during the conference will include the following sessions. We will also appropriately incorporate breaks and social activities into the final schedule.

Introduction (1 hour): We will provide a brief introduction to the workshop outlining planned activities, goals and themes of the workshop. We also plan to run a quick ice-breaking round to get to know the participants.

Position Paper Discussion (1.5 hour): In the synchronous session, participants will share their position papers with the audience. We plan on allocating time for selected papers, and the presentations

¹<https://sadworkshop.wordpress.com/>

²<https://sites.google.com/view/crowdbias>

³<https://research.yandex.com/workshops/crowd/neurips-2020>

⁴<http://eval.how/dew2020/>

will be organised under workshop themes. Each presenter will get 3 minutes to present their work followed by 7 minutes of questions from the audience. We want to prioritise the opportunity for authors to obtain feedback from the audience. We will also share the position papers with participants prior to the workshop and encourage them to read them beforehand.

Crowd Bias Challenge Recap (1 hour): The challenge outcomes will be presented and discussed during the workshop. We will invite the leading three teams to discuss their solutions briefly.

Blue sky session - beyond the crowd (1 hour): This ideation session will be facilitated by an invited expert outside the crowdsourcing research domain to explore challenging future research directions. We will probe participants to discuss more broad research questions around biases and crowd data. For example, how can we scale crowdsourcing platforms to wider population groups while preserving bias considerations? We plan to conduct small group discussions under four themes.

Closing (30 mins): We will use this session for summarising the workshop output and obtaining feedback from participants regarding possible future activities. We will also facilitate follow-up conversations after formally concluding the workshop.

2.5 Post Workshop Activities

We will maintain the slack workspace to facilitate follow up conversations. We also plan to publish the outcome of the crowd bias challenge and anticipate that other successful participant teams will extend their work to similar outputs. With permission of participants, we hope to publish workshop proceedings through CEUR (<http://ceurws.org>). In addition, based on workshop discussions, we intend to document a list of biases in crowdsourced data, their sources and suggested methods to mitigate them during or after labelling.

2.6 Virtual Setup and Participants

We plan to recruit 20-30 participants for the workshop. Participants will be required to either submit a position paper or take part in the workshop challenge. Our workshop will align with the interests of researchers and practitioners in crowdsourcing, CSCW, HCI and IR who explore biases in crowd data. The workshop will also appeal to individuals who use crowdsourcing for applications in broader areas of machine learning, social science and data science. We will promote our workshop, challenge and the position paper call via online mailing lists, social media and forum. The workshop will also be actively promoted through research groups, and centres organisers are affiliated with (e.g., multidisciplinary ARC Centre of Excellence on Automated Decision Making and Society⁵). We plan to utilise the following tools to deliver the virtual workshop.

Workshop Website: Workshop website will publish all public information including the call for position papers. *Slack Workspace:* We will setup a slack workspace dedicated to the workshop to enable communication among participants, particularly during the pre-workshop activities and to support followup conversations. We will also use the workspace to share the crowd bias challenge material and accepted position papers prior to the workshop. *Virtual Work-*

shop Video Conferencing Platform: We will use Zoom or Microsoft Teams to run the synchronous session of the workshop. Both platforms have features that allow us to create breakout rooms, share content and secure the session.

In addition, we plan to incorporate regular interactive polls to get continuous feedback from the participants and maximise the engagement.

3 ORGANISERS

Our team consists of scholars and industry leaders working in and across CSCW, HCI, IR and Crowdsourcing. In addition to a strong record of being part of conference organising activities, the team also have prior experience with related workshops [24, 29] and running challenges [25, 30].

Danula Hettiachchi is a Research Fellow at the ARC Centre of Excellence on Automated Decision Making and Society and RMIT School of Computing where he research user biases when interacting with automated systems. His doctoral research examined task assignment in crowdsourcing [13, 14].

Mark Sanderson is a Professor of information retrieval (i.e. search engines). He has published extensively in the areas of evaluation of search engines [30], user interaction with search, and conversational searching systems. Mark is a Chief Investigator on the \$32 million Australian Government Centre of Excellence, Automated Decision Making and Society (ADMS). He is a visiting professor at the National Institute of Informatics in Tokyo.

Jorge Goncalves is a Senior Lecturer in the School of Computing and Information Systems at the University of Melbourne. He has conducted extensive research on improving crowd data quality [10, 14, 15], and bringing crowdsourcing beyond the desktop by using ubiquitous technologies [11, 12]. He has also served as Workshops Co-Chair for CHI'19 and CHI'20.

Simo Hosio is an Associate Professor and the Principal Investigator of the Crowd Computing Research Group at the Center for Ubiquitous Computing, University of oulu, Finland. His research focuses on social computing, crowd-powered solutions for digital health [16], and crowdsourced creativity.

Gabriella Kazai is a Principal Applied Scientist at Microsoft, focusing on offline evaluation, crowdsourcing, and metric development for various search scenarios, including organic web search, news, autosuggestions, and covering aspects from relevance to source credibility. She co-organised a number of evaluation initiatives, including the TREC crowdsourcing track [25, 36] and INEX, and currently serves on the Steering Committee of the AAAI Human Computation and Crowdsourcing (HCOMP) conference and as PC chair for SIGIR 2022. Her research interests include information retrieval evaluation [30], human computation, gamification, recommender systems, and information seeking behaviour.

Matthew Lease is an Associate Professor in the School of Information at UT Austin and an Amazon Scholar in Amazon's Human-in-the-Loop services. He has conducted extensive research in crowdsourcing and human computation for a decade [13, 22, 24, 26, 30, 36] and currently serves as Steering Committee co-chair of the AAAI Human Computation and Crowdsourcing (HCOMP) conference.

⁵<https://www.admscentre.org.au/>

Mike Schaekermann [13, 32, 33] is an Applied Scientist in Amazon's Human-in-the-Loop services. His dissertation work in crowdsourcing and human computation was recently honoured with the 2020 *Distinguished Dissertation Award* from CS-Can[Info-Can, Canada's key computer science professional society.

Emine Yilmaz [21, 24, 26, 35] is a Professor and Turing Fellow at University College London, Department of Computer Science. She also works as an Amazon Scholar with Amazon Alexa Shopping. She has been working on modelling and evaluating annotator quality, user modelling and evaluating bias in information retrieval systems.

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